ANUJ SHARMA

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About

Final-year CSE (Data Science) student and Backend & AI Engineer who ships AI inside production-grade software: self-hosted speech-to-text (STT) and text-to-speech (TTS) models, LLM agents, RAG and vector search, and fine-tuned PyTorch vision models, served through FastAPI and Spring Boot. Focused on verification over guesswork, concurrency-safe data and adversarial testing, backed by strong DSA and system design fundamentals.

Education

Vellore Institute of Technology, Vellore 2023 – 2027
B.Tech. in Computer Science and Engineering (Data Science); CGPA: 8.34/10
DAV Public School | Senior Secondary (CBSE): 92% 2022
St. Vincent's High and Technical School | Secondary (ICSE): 85% 2020

Experience

FOMO Entertainment Pvt. Ltd. Oct 2025 – May 2026
Backend & AI Engineer *Remote*

- Built an LLM-based semantic verification agent on Google ADK that reads a parsed .eml as transcript context alongside the seller's listing and compares event, artist, venue, date and buyer identity field by field, catching a valid signature on a ticket for the wrong event
- Engineered the ingestion and verification stack in Spring Boot against SMTP and MIME specifications: parsing headers and body parts, validating the cryptographic signature and sender domain, and proving the seller forwarded the original, unaltered email
- Authored and ran the backend test suite over adversarial cases (mismatched events, forwarded chains, tampered bodies, replayed tickets), validating that genuine and fraudulent listings were classified correctly before rollout

Projects

CliniQ | Multilingual Voice-Agent Booking & Clinic Management SaaS GitHub
Stack: Python, PyTorch, FastAPI, AI4Bharat IndicConformer, Whisper, Meta MMS-TTS, Java 21, Spring Boot, PostgreSQL, Redis, RabbitMQ, React, TypeScript, Docker

- **STT:** Built the speech-to-text layer of a voice booking agent for Hindi, Tamil and English on self-hosted PyTorch models (AI4Bharat IndicConformer, Whisper), so patient audio never leaves clinic hardware
- **TTS:** Added text-to-speech with Meta MMS-TTS so the agent answers aloud, closing a fully self-hosted speech loop (STT, intent handling, TTS) with no third-party speech API
- Wrote intent modules that handle code-switched speech (Hindi, Tamil and English mixed together) and refuse symptom questions, keeping the agent strictly inside booking scope
- Guaranteed a slot is never booked twice using SELECT FOR UPDATE row locks and a partial unique index, proven by a Testcontainers race test and a load test: 96 concurrent attempts on 32 slots produced exactly 32 bookings and 64 clean 409 rejections (p95 261 ms at 100 concurrent clients)
- Shipped Razorpay payments with webhook signature verification, JWT auth with RBAC and rate limiting, timed slot holds and RabbitMQ reminders, backed by 350+ automated tests (JUnit, Vitest, Playwright E2E) gated by CI; removed a Redis cache that load testing showed was 6.4x slower than the query it replaced

SkyNetra | Satellite Image Detection, Visual Search & Retrieval System GitHub
Stack: Python, PyTorch, YOLOv8, ResNet50, FAISS, Scikit-learn, OpenCV, FastAPI, Streamlit

- Fine-tuned YOLOv8 in PyTorch on a 14-class satellite dataset with automatic YOLO-format label generation, and designed a hybrid semi-supervised pipeline: detections above 60% confidence keep their supervised label, while uncertain regions are routed to K-Means clustering (87% purity) instead of being forced into a wrong class
- Built visual similarity search on 2048-dim ResNet50 embeddings indexed in FAISS (cosine similarity), returning the top-5 matches in sub-100 ms across 1,000+ images, served through a FastAPI /analyze endpoint with a Streamlit UI

Technical Skills

AI & ML: PyTorch, Transformers (Hugging Face), LLM API Integration, RAG, LangChain, LangGraph, Google ADK, Vector Search (FAISS), Speech AI (STT/ASR, TTS), Prompt Engineering, YOLOv8, Scikit-learn
Languages: Java, Python, C++, SQL, R, JavaScript, TypeScript
Backend: Spring Boot, FastAPI, REST APIs, RabbitMQ, JWT Auth & RBAC, React
Databases: PostgreSQL, MySQL, MongoDB, Redis
DevOps & Tools: Docker, GitHub Actions CI/CD, Testcontainers, Playwright, Git, Linux, AWS (EC2, S3)
Data: Pandas, NumPy, Matplotlib, Seaborn, EDA, Feature Engineering

Achievements & Certifications

- Oracle Certified Generative AI Professional (Oracle, 2024); AI Engineer Core Track: LLM Engineering, RAG, QLoRA and Agents (Udemy)
- LeetCode: 100+ days of consistent problem solving; completed the SQL 50 study plan

Leadership

Operations & Finance Executive | E-Cell VIT Aug 2024 – Apr 2025

- Analysed budgets across 10+ events and drove changes that cut operational costs by 15% and execution delays by 20%

Student Coordinator | Riviera Fest 2026, VIT Jan 2026 – Mar 2026

- Coordinated 20+ members for VIT's flagship fest (30,000+ attendees); led outreach reaching 10,000+ students